Final exam questions will include questions and calculations similar to ones you perform in your lab reports .

The following is a selection of examples of calculations questions from previous CH316 exams. These are provided to help you become familiar with the format of such questions on the final exam. This is NOT representative of the full list of concepts and calculations covered in the lab or of the proportion of questions from the covered concepts.

1. Consider your lab experiences and any relevant equations to fill in the blank with the appropriate answers.

A. A sample was determined to contain 1.03356 μg analyte/mL with a 95% confidence limit of 0.000821 μg /mL. Report the concentration to the correct number of significant figures based on the confidence limit. _____________

B. In a mixture of two solutes where one is polar (A) and the other is non-polar (B), which solute will elute first on a C18 column, A or B? _________

C. During quantitative chemical analysis experiments, you are usually asked to record replicate measurements. What type of error is minimized by repeat measurement, random or systematic error? _______________

D. During the analysis of fluoride content in mouthwash using an ion-selective electrode, it was important to adjust the pH of the solution using the TISAB buffer. If the pH of the solution is too high would the determined fluoride content be lower or higher than the true value? ______________

E. Which calibration curve is expected to show a non-zero and positive y-intercept, external calibration curve or standard addition calibration curve? ______________

2. A calibrated curve based on a set of known standards was used to measure the μg RNA in an unknown. The responses (absorbances) from the standards and unknown were measured and reported here. The first two lines from the Linest output are also provided here.

	μg RNA	Absorbance (260nm)
Standard1	0	0.003
Standard2	11.02	0.24
Standard3	21.95	0.44
Standard4	34.15	0.71
Unknown – run 1		**0.33**
Unknown – run 2		**0.31**

Linest (first two lines)	
0.0205	0.00454
0.00051	0.0107

The expression for calculating the standard error in the μg RNA in the unknown is given here

$$s_c = \frac{s_r}{|m|}\sqrt{\frac{1}{M} + \frac{1}{N} + \left(\frac{(\overline{y}_u - \overline{y})s_m}{ms_r}\right)^2}$$

What numerical values of $\overline{y}$, $\overline{y_u}$, M and N should be used in this equation?

$\overline{y}$	
$\overline{y_u}$	
N:	
M:	

3. Consider the provided data from the calibration of a 10mL volumetric pipet. The volumes of dispensed water were calculated from the mass using the density of water at the lab temperature. The average and standard deviation were calculated using the Average() and stdev() functions in Excel.

	Mass of dispensed water (g)	Actual volume of dispensed water (mL)
Trial 1	9.9748	9.994833 (note values are copied
Trial 2	9.9801	10.00014 over from Excel witout
Trial 3	9.9795	9.999542 adjustment of
	Average	9.99817 significant figures)
	Standard deviation	0.002908

a. Report the 95% confidence interval in the <u>average volume</u> delivered by this pipet. Report the average volume delivered to the correct number of significant figures based on the confidence limit.

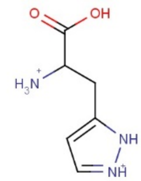
b. Is the determined average volume significantly different than the label volume (10mL) at the 95% confident level? Support your answer with the appropriate calculations/equations for credit.

4. Consider the titration of 25.00 mL of a solution of unknown KHP concentration with a 0.1295 M NaOH secondary standard solution. Phenolphthalein is used to identify the endpoint of the titration. Three titration attempts were performed and the average volume of NaOH needed to reach the endpoint is found to be 21.25 mL

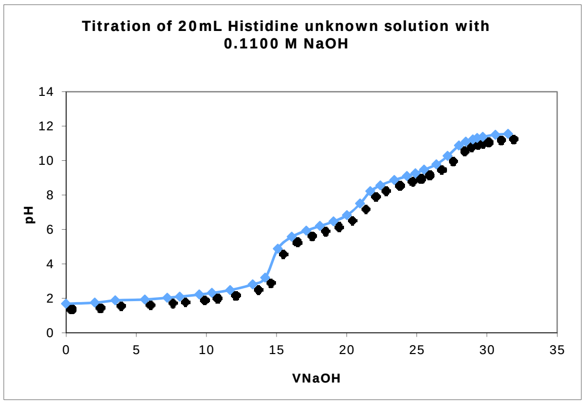
a. Calculate the molarity of the KHP solution. Report the answer to 4 significant figures.

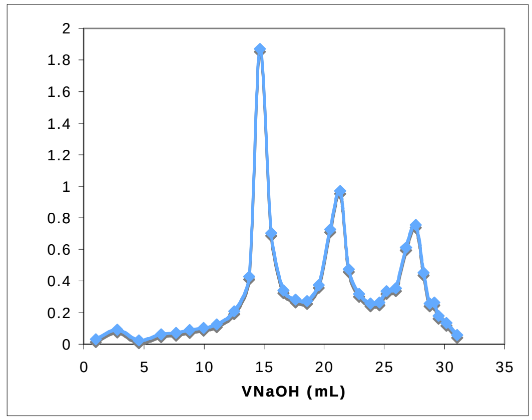
b. If the buret was rinsed with DI water and was not conditioned with the NaOH solution before filling with NaOH, do you expect the calculated concentration of KHP from this titration to be higher or lower than the expected value? Very briefly explain how you reached your answer for credit.

5. Use the provided titration curve and first derivative plot to answer the following questions about the titrated L-Histidine hydrochloride solution.



(the COOH proton is the most acidic, then the -NH^{1+}, then the -$NH3^{+}$protons)





a. Estimate the pKa_2 of Histidine. Explain/mark on the curve above how you reached your answer.

b. What do the three maxima correspond to in the plot (right) of $\Delta pH/\Delta V$ vs V_{NaOH}?

6. Cell potential readings in mV were recorded when a F^- ion-selective electrode was immersed in a series of standard solutions to which TISAB buffer was already added. The plot of E (in mV) vs log [F^-] was linear. Regression statistics were determined using LINEST function and excel (given here).

-54.6192	50.4437
0.51776	1.37588
0.99955	1.36997
11128.03	5
20885.50	9.38418

a. A mouthwash sample with unknown F^- concentration was also analyzed after addition of TISAB buffer and gave a reading of 210.5 mV.

The s_c expression given in the equation sheet was used to determine the standard error in the log[F^-] for the unknown and returned 0.0250. Calculate the 95% confidence interval in the molar concentration [F^-] in the unknown. Report the [F-] in the unknown to the correct number of significant figures based on the confidence limit.